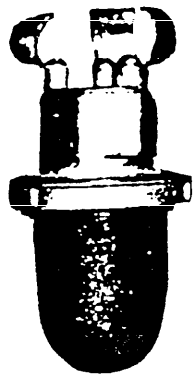
INCH-POUND
MIL-F-16377/27B(SH)
7 October 1988
SUPERSEDING
MIL-F-16377/27A(SH)
9 May 1983

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING
(B) 50 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 92.2 AND 93.2

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-F-16377.



SYMBOL 92.2 OR 93.2



SYMBOL 92.2 OR 93.2 WITH BULKHEAD MOUNTING BRACKET

SH 13231839

FIGURE 1. Fixtures, incandescent, symbols 92.2 and 93.2.

NOTES:
1. All dimensions in inches.
2. For tolerances, other than specified herein, see MIL-F-16377.

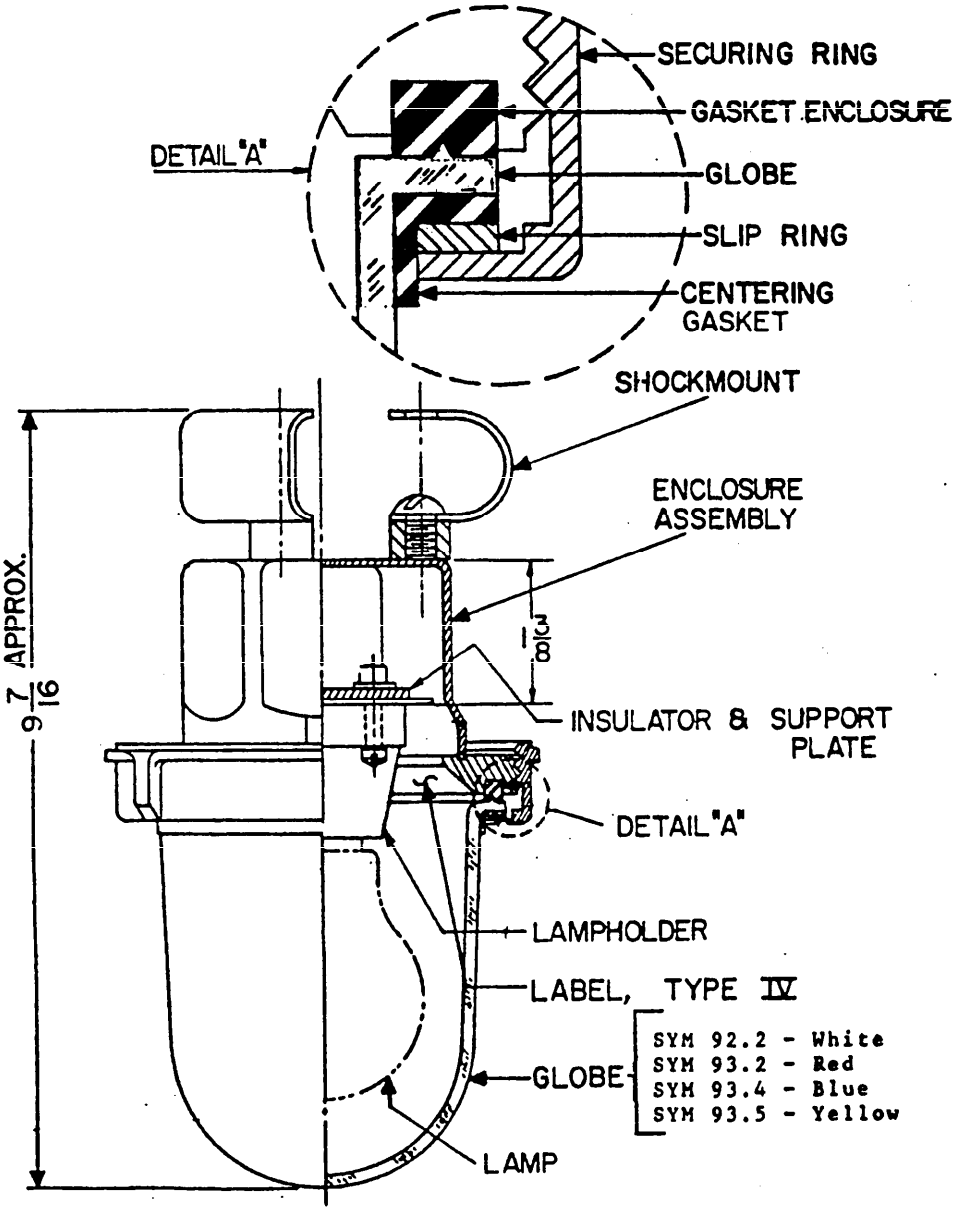
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FSC 6210

AMSC N/A

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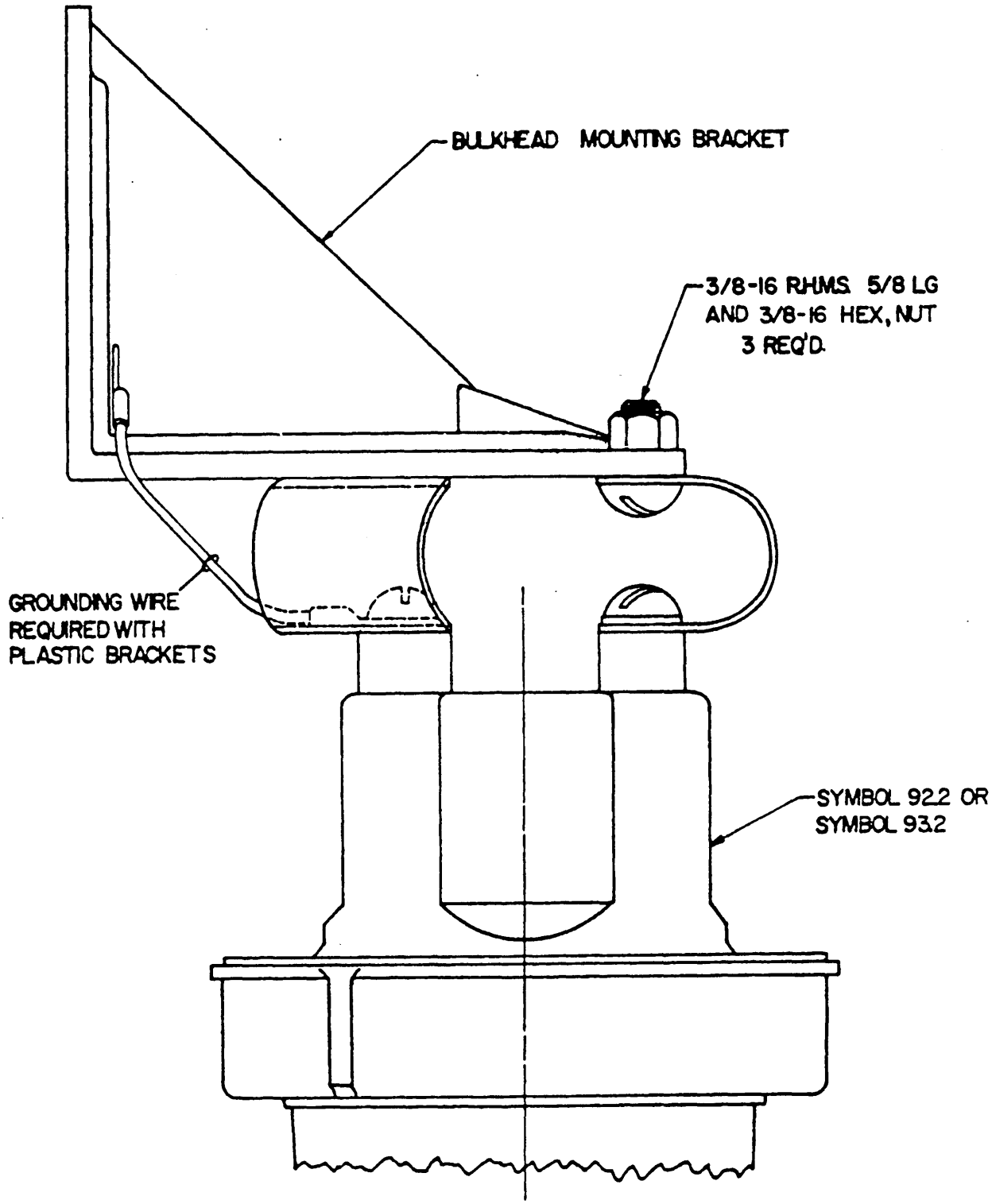
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SH 8751B

FIGURE 2. Fixture assembly.

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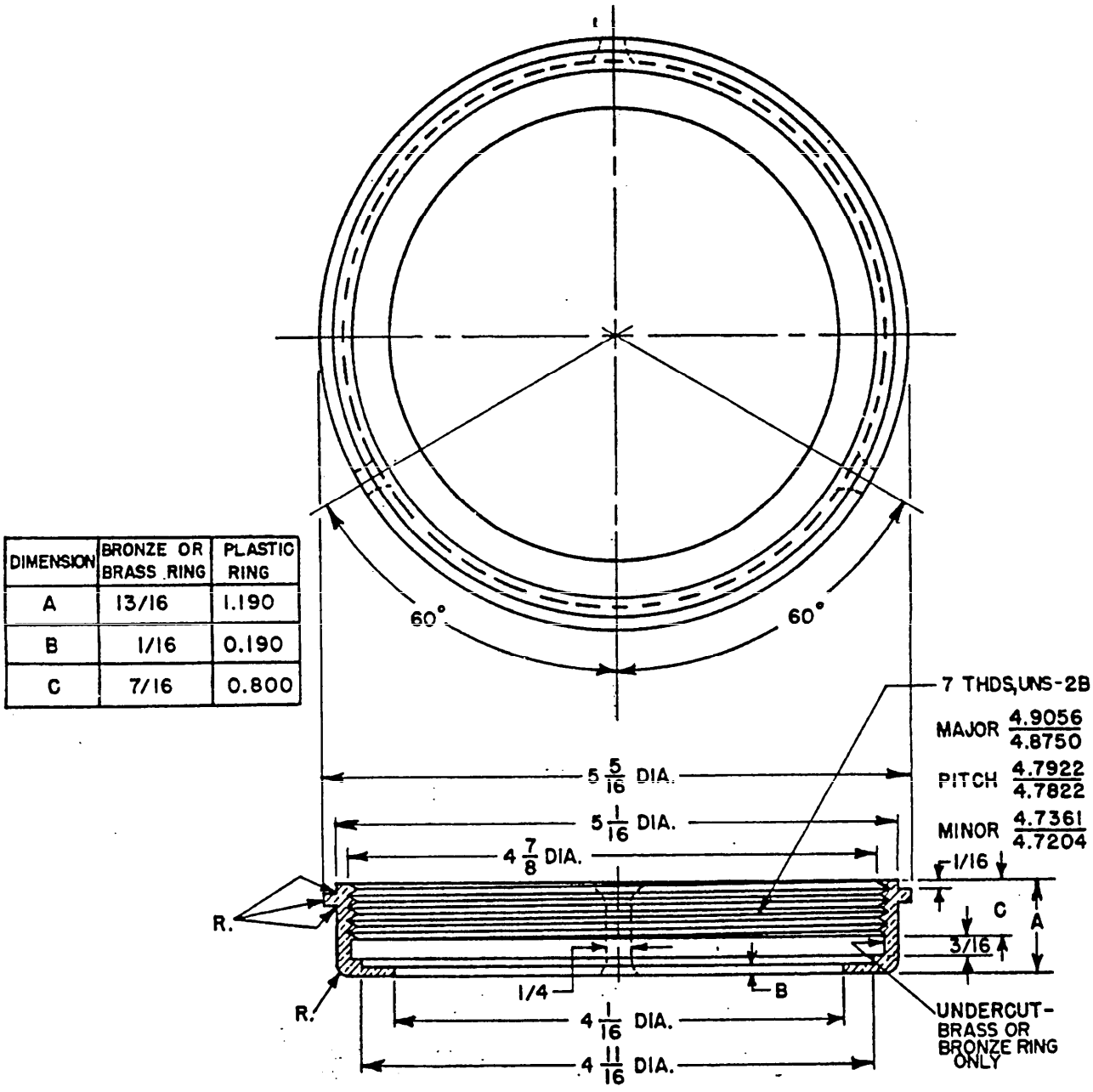


SH 13231835

FIGURE 3. Fixture assembly for bulkhead mounting.

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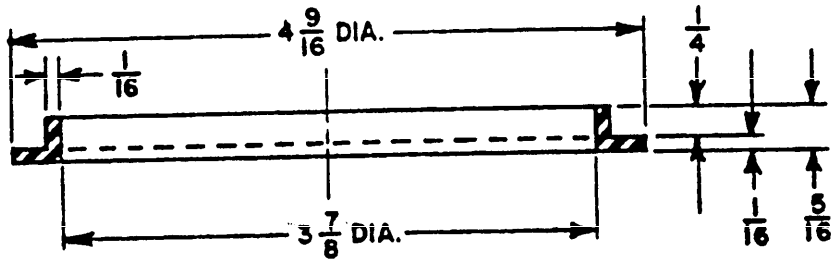
DIMENSION	BRONZE OR BRASS RING	PLASTIC RING
A	13/16	1.190
B	1/16	0.190
C	7/16	0.800



SH 10025A

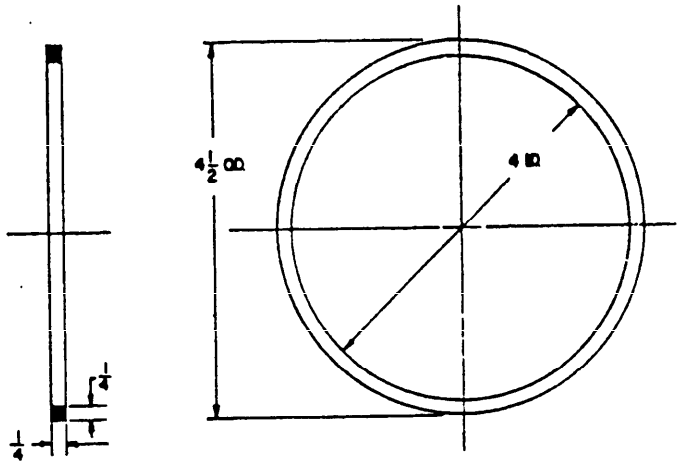
FIGURE 4. Securing ring.

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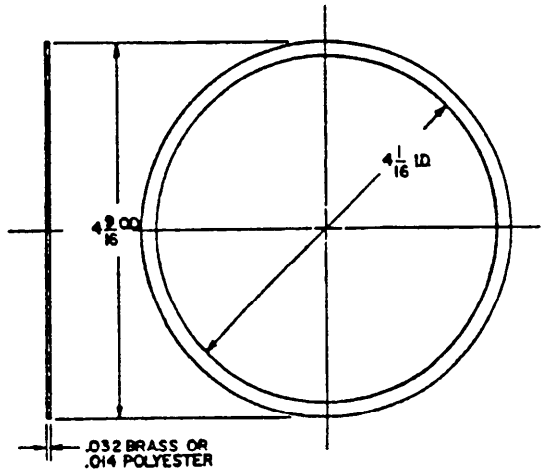
SH 10026

FIGURE 5. Globe centering gasket.



SH 13231836

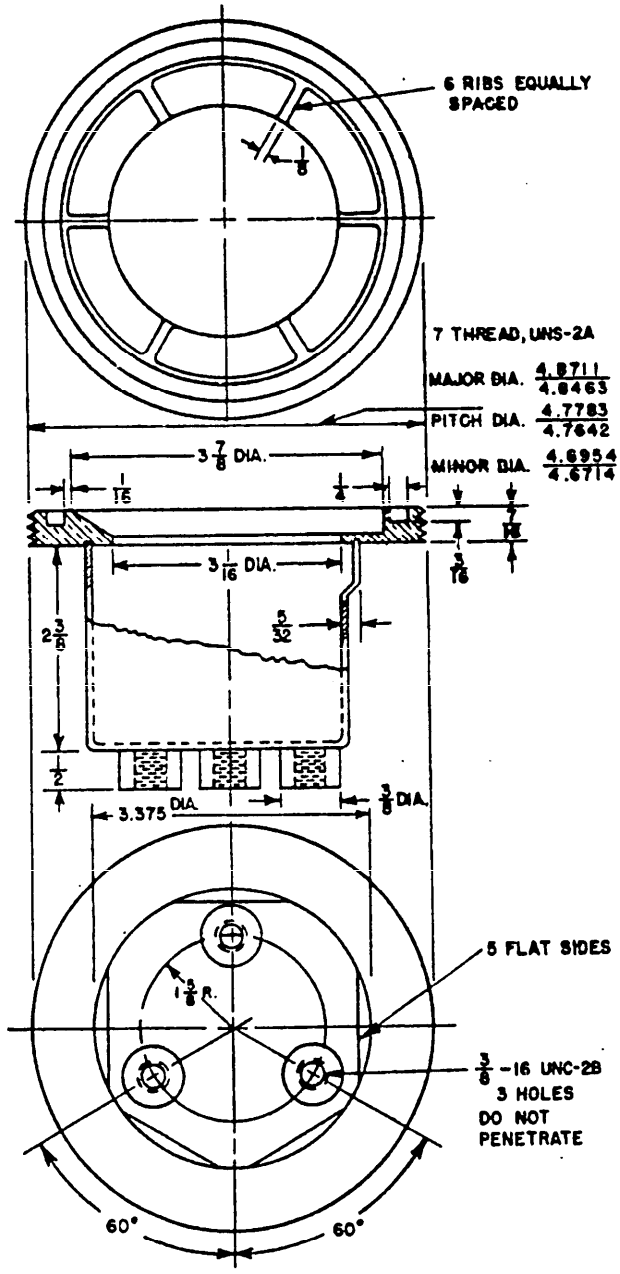
FIGURE 6. Enclosure gasket.



SH 13231837

FIGURE 7. Slip ring.

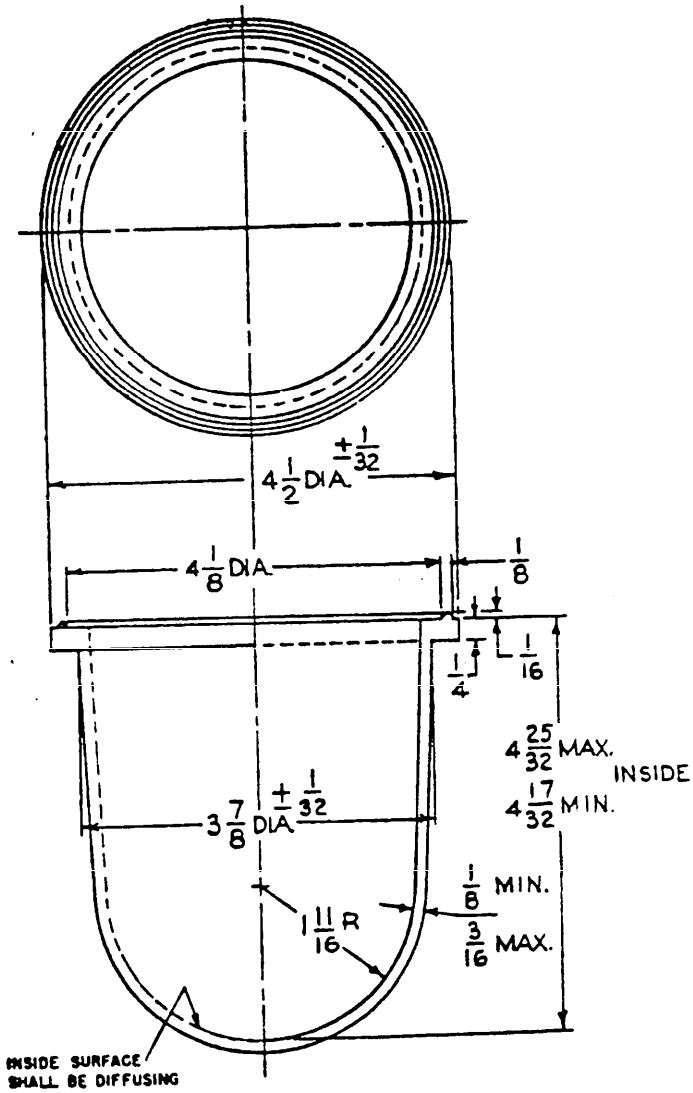
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SH 10027A

FIGURE 8. Enclosure assembly.

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Material - Tempered glass
Color:

Symbol 92.2 - Clear transparent/diffusing.
Symbol 93.2 - Red transparent/diffusing.
Symbol 93.4 - Blue transparent/diffusing
Symbol 93.5 - Yellow transparent/diffusing

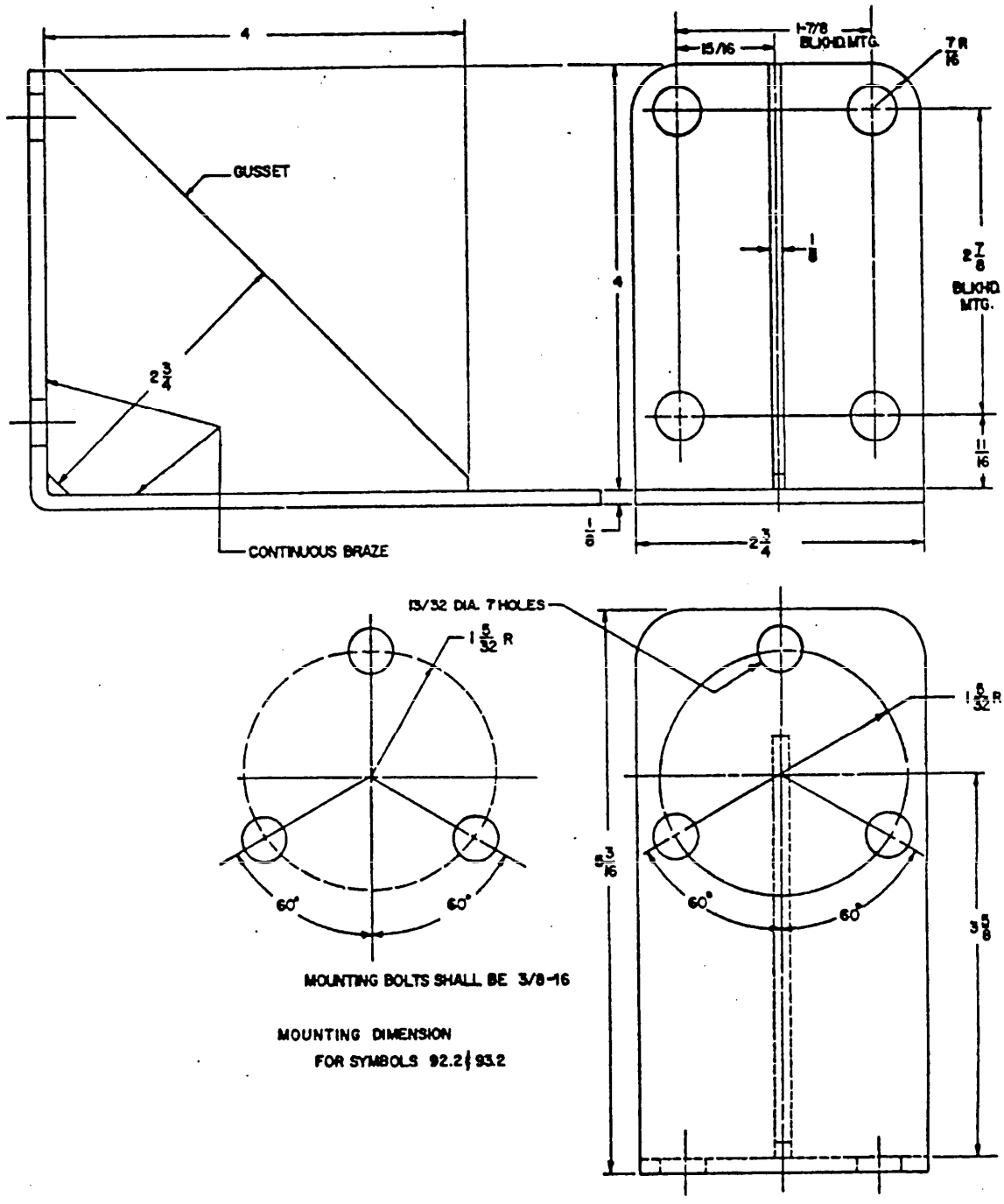
Transmittance (symbol 92.2):
Clear transparent/diffusing - 80 percent minimum.

SH 10000

FIGURE 9. Globe.

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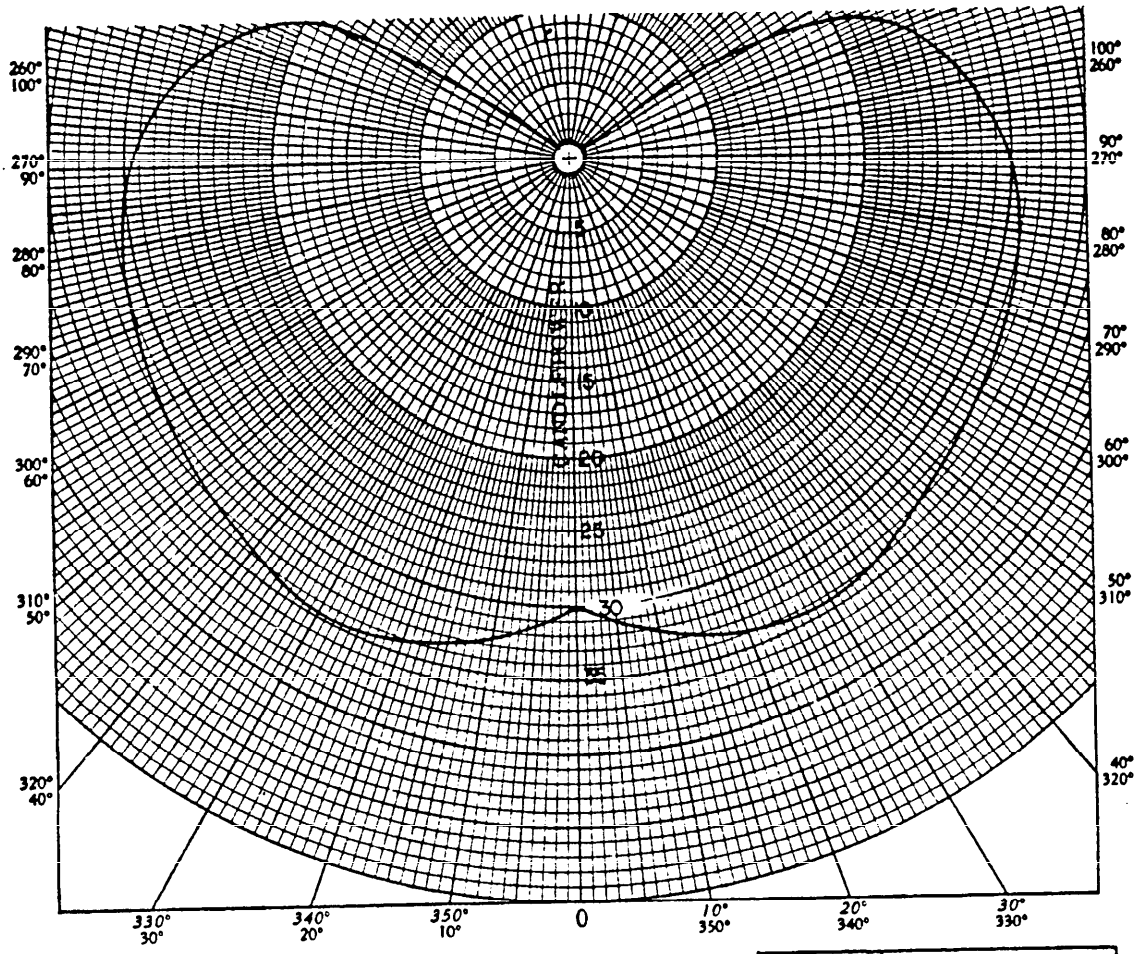
Type Rapid Start.



SH 8753A

FIGURE 10. Mounting dimensions and bracket for bulkhead mounting (metal bracket only).

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ANGLE	CANDLEPOWER SYMBOL 92.2
0	30
5	32
10	32.5
15	33
20	33.5
25	34
30	34
35	34
40	33
45	32
50	32
55	32
60	31.5

ANGLE	CANDLEPOWER SYMBOL 92.2
65	31
70	31
75	31
80	30.5
85	30
90	30
95	28
100	27.5
105	27
110	25
115	23
120	12.5
125	2

Photometric test data:

Distance: 10 feet.
Voltage: 120 volts.
Ambient temperature: 76°F.
Lamp: 50 watts, rough service conforming to MS15586-2, furnished by laboratory for test, shall be used.

SH 13231838

FIGURE 11. Candlepower distribution curve.

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REQUIREMENTS:

Type: II
Class: 2
Symbol 92.2: Fixture for use with one 50-watt white incandescent lamp and white (clear transparent) diffusing globe.
Symbol 93.2: Fixture for use with one 50-watt white incandescent lamp and red (transparent diffusing) globe.
Symbol 93.4: Fixture for use with one 50-watt white incandescent lamp and blue (transparent diffusing) globe.
Symbol 93.5: Fixture for use with one 50-watt white incandescent lamp and yellow (transparent diffusing) globe.

Design:

Electrical rating: 120 volts, 50-watts.
Dimensions and configuration: See figures 2 through 9.
Enclosure assembly (includes box, threaded flange, tapped pads, and support plate): See figure 8. Material shall be brass, or a gray plastic, type MAI-60 in accordance with MIL-M-14.
Insulator for brass box assembly only: Plastic.
Gasket: Rubber, 1/4-inch square cross section by 4-inch inside diameter (id) by 4-1/2-inch outisde diameter (od).
Lampholder: MIL-L-970/6, two contact.
Shockmounts: Brass, MIL-F-16377/41-012, or MIL-F-16377/41-015.
Centering gasket: Molded rubber, see figure 5.
Globes:
Symbol 92.2: White (clear transparent) diffusing. See figure 9.
Symbol 93.2: Red transparent diffusing. See figure 9.
Symbol 93.4: Blue transparent diffusing. See figure 9.
Symbol 93.5: Yellow transparent diffusing. See figure 9.
Slip ring: Brass, 4-9/16-inch od by 4-1/16-inch od by 1/32-inch thick.
Securing ring: Cast brass, bronze, or gray plastic, type MAI-60 in accordance with MIL-M-14.
Bulkhead mounting bracket: When specified in the contract, a brass, corrosion-resistant steel, or gray plastic type MAI-60 mounting bracket in accordance with MIL-M-14. The metal brackets shall be as shown on figure 3. Plastic brackets shall have mounting dimensions as shown on figure 3. Thickness of material shall be as necessary to meet the requirements specified herein. Metal inserts are required on the mounting holes of plastic brackets.
Lamp: Fixtures shall accommodate a 50-watt white lamp conforming to MS15586-2. Lamp is not furnished.
Distribution curves (Symbol 92.2): Candlepower (see figure 11).
Weight (Symbols 92.2 and 93.2): 2 pounds, 13 ounces max.
Labels: Labels shall be furnished in accordance with MIL-F-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be:

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"INSTALL 50-WATT LAMP ONLY.
LAMP: NSN6240-00-143-3070
WHITE GLOBE: NSN6210-00-702-5773
RED GLOBE: NSN6210-00-914-4152"
Identification plate: Brass, stamped or engraved.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Globes:	See table II.
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-F-16377.
Examination for preparation of delivery:	See MIL-F-16377.
Quality conformance tests:	See MIL-F-16377.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement	Test	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE II. First article and comparison inspection for globes.

Tests	Requirement	Test	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectivenesss	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only

General information: (a) Assignment of Military part numbers and general information is shown in table III.
(b) Fixtures symbols 92.2 and 93.2 may be adapted for bulkhead mounting by using the bracket shown on figure 3.

TABLE III. Part number and general information.

Military part number M16377/27-	Item name	National Stock No. (NSN)	Supersedes part number
92.2	Complete fixture, Symbol 92.2 consisting of: body and white (clear transparent diffusing) globe	6210-00-299-7703 6210-00-702-5773	9000-S6401-74218 Symbol 92.2
93.2	Complete fixture, Symbol 93.2 consisting of: body and red transparent diffusing globe	6210-00-299-7703 6210-00-914-4152	9000-S6401-74218 Symbol 93.2
001	White (clear transparent diffusing) globe	6210-00-702-5773	MS17802-20
002	Red (transparent diffusing) globe	6210-00-914-4152	MS17802-21
003	Mounting bracket	5340-00-391-1073	
004	Body (complete less globe and mounting bracket)	6210-00-299-7703	

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TABLE III. Part number and general information. - Continued

Military part number M16377/27-	Item name	National Stock No. (NSN)	Supersedes part number
005	Securing ring	6210-00-399-6207	
006	Blue (transparent/diffusing) globe		
007	Yellow (transparent/diffusing) globe		

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-16377/27B(SH)	FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING 50 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 92.2 AND 93.2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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